Experiment No 08 :-

Theory

- phenol - water phase diagram
- phenol composition = $\frac{m_p}{m_p + m_\omega}$ m_p = m_ω = $\overline{\omega}$ units

Observations

Table No 01:- Temperature at which turbidity disappear and appear

Volume of distilled water added (± 0.05) mL	Temperature of turbidity disapper (± 0.5) /°C	Temperature of turbidity ~~disappear~~ (± 0.5) /°C	Average Temperature
1.00	T_1	T_2	$(T_1 + T_2)/2$
2.00			
3.00			
4.00			
5.00			
⋮			
12.00			

Calculation.

Density of distilled water $= 1\ g\ cm^{-3}$

Density of phenol $= 1.07\ g\ cm^{-3}$

mass of phenol $= V . \rho$

$= 1.07\ g\ cm^{-3} \times 2 cm^3$

$= 2.14\ g$

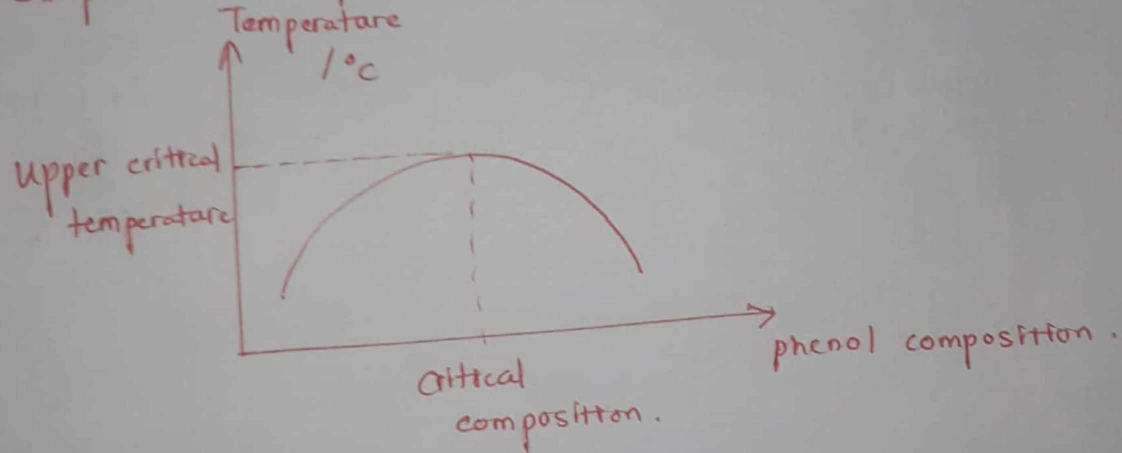
Table No 02 :- percentage of phenol ~~composition~~.

Volume of distilled water (± 0.05) cm^3	Mass of water ($\pm 1g$)	total mass ($\pm 1g$)	percentage of phenol
1.00	1.0	1.0 + 2.14	$(2.14/3.14)$ ↑ $m_p/(m_p + m_w)$
2.00	2.0	2.0 + 2.14	
⋮	⋮	⋮	
12.00	12.0	12.0 + 2.14	

Table No 03 :- Values for phase diagram

Average temperature (±0.5) /°C	composition of phenol

Graph No 01 :- phase diagram for phenol - water system.



conclusion :-

upper critical temperature is ______ °C and the critical composition is ______ %

Discussion :-

- Define the phenol - water system.
- phase diagram of phenol - water system
- upper critical temperature.
- Error of the experiment

Reference :-